Creating a Bivariate Thematic Map
Luis Edgar Vargas Ramirez, 11/13/2024

The aim of this report is to identify and analyze patterns in the spatial distribution of the "Other races" group population in relation to household income levels across various areas. Specifically, the analysis focuses on the percentage of people in each area who identify as Native Hawaiian and Other Pacific Islander, American Indian and Alaska Native, some other race alone, or two or more races —referred to collectively as the "Other races" group. The second variable examined is the estimated median household income in 2022. By using a bivariate map, this report seeks to address the analytical question: *Where are the areas with a high concentration of the "Other races" group and low median household income in the City of New York?*

The relationship between the percentage of the other races group population and household income levels provides insight into the socioeconomic disparities affecting diverse racial groups. These two variables—racial identity and income—are often interconnected due to historical, social, and economic factors that shape opportunities and outcomes. The other races group represents communities that may face unique barriers to socioeconomic advancement, including limited access to quality education, healthcare, and employment opportunities.

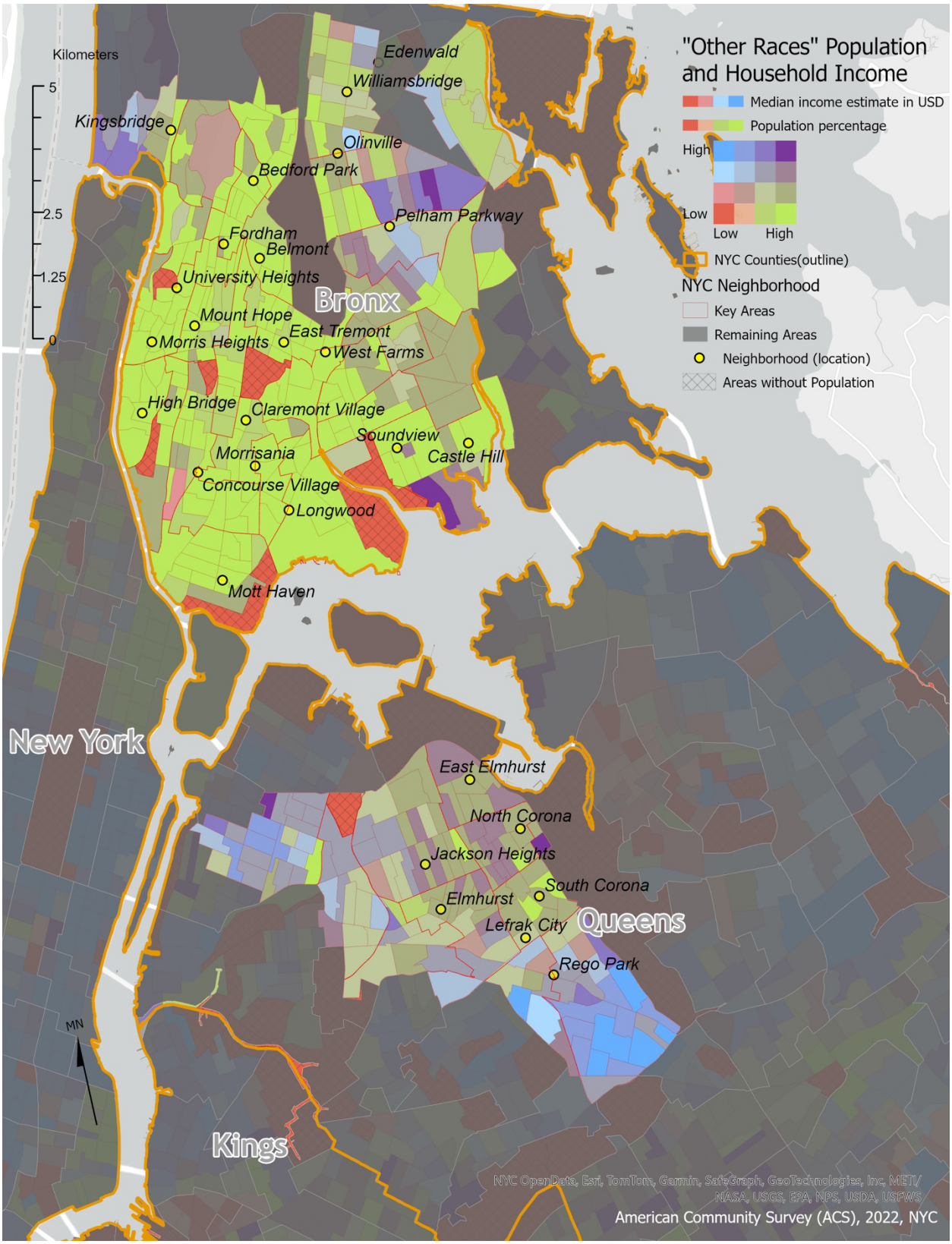
Income disparities in areas with a higher concentration of the other races group may reflect broader trends of economic marginalization and inequality. By examining these variables together, we can identify patterns that indicate areas where these communities experience lower household income levels, suggesting a need for targeted social programs, resources, and policy interventions. Additionally, a spatial analysis of these patterns may reveal how neighborhood factors, such as access to economic resources and quality services, influence both income and racial diversity within an area.

The data distribution for both variables was analyzed, finding that both distributions were right skewed, the mean median alignment was off to the right by moderate amounts(about 5 for the population percentage and 2875 for the median income ), and the median household income is marginally closer to normality than the race group population. The intention to use the corner model along with these findings helped make the decision that the breaks for the manual interval classification scheme were going to be the median and upper and lower quartiles. This is because median represents the central point of the data regardless of skew, and Q1 and Q3 reflect the spread without being distorted by skewness.

The focal model used in the analysis was the corner model. As mentioned in the paper, Operationalizing Trumbo's Principles of Bivariate Choropleth Mapping by Strode HL(2019), this model highlights the distinct corners while minimizing the interior, which is done by using complementary diverging color schemes. The first diverging color scheme used was a crimson red for the lowest part of the population percentage and lime green for the highest part. For the household income, crimson red was used for the lowest part and azure blue for the highest part of the scheme. These three colors along with purple complete the corners of the model. The color scheme was chosen with the objective of this model in mind. According to the paper, the goal of the corner model is to allow quick and easy identification of areas with very low or very high values of either variable without distraction from areas with intermediate values. The four corner value colors are distinct from each other and allow discrete areas to be identified. The twelve intermediate colors were color variables that were either toned down, more transparent, or both, so that they would not be the main focus of the reader's attention.

A 4x4 thematic mapping grid was selected because this grid allows for a more detailed analysis due to how many gradients there are in the map. The main four corner colors represent the different high and low relationships between the variables:

*Figure 1, ' NYC Bivariate map showing areas with a high percentage of "other races" population with low median household income, to reveal patterns of demographic and economic disparities'.*



- Crimson red (Low x Low): low concentration of the “Other races” population with low median household income.
- Lime green (High x Low): high concentration of the “Other races” population with low median household income.
- Azure blue (Low x High): low concentration of the “Other races” population with high median household income.
- Purple (High x High): high concentration of the “Other races” population with high median household income.

The central colors in between the main corners follow a divergent color scheme, indicating intermediate values where one variable is moderate while the other is high or low. For example, one shade may represent a low concentration of the other races population with a moderate median household income, while another could represent a moderate concentration of the target population with a high median household income. These 4 main divergent schemes help minimize the values in between the variable range while at the same time highlight the extremes.

The last four middle values in the center of the grid diagram are a blend of the extreme diagonal corner values. They follow the same divergent scheme progression, but going from Low x Low to High x High, as well as from Low x High to High x Low. This section is also intended to be minimized in order to direct attention to the extremes.

At first sight the map highlights two specific areas: one in the Bronx and another in Queens. Within both of those areas, two distinct clusters are highlighted. The one in the Bronx has a larger number of areas with high concentration of the other races population and low median household income. Meanwhile the one in Queens is smaller and more dispersed. Both clusters exemplify the corner model approach to emphasize the extremes, but the one in Queens is particularly exaggerated because the corner areas are surrounded by a higher number of median areas.

The vast majority of the areas in the Bronx display a medium-high to high population concentration with a medium-low to low income association. The West and Southwest Bronx have the most areas with a high population concentration and a low median household income, specifically in neighborhoods like Morris Heights, Longwood, University Heights, and Fordham. A transition to moderate-high to high income associations can be observed from the southwest part of the Bronx to the northeastern part, in neighborhoods like Pelham Parkway, Olinville and Williamsbridge. These neighborhoods still have areas with a high population concentration and a low median income relationship, but we now see more high population with mid-high median income relationship areas, including one unique high population with high income area.

The cluster in Queens is smaller than the one in the Bronx, and highlights four high population with low median income tracts that are located in the neighborhoods of South Corona, North Corona and Jackson Heights. These areas are surrounded by transitioning areas that have high population concentration and a moderate median income, with a unique high population with high median income area in the North Corona neighborhood.

According to the First Law of Geography, which states that things that are related are more related the closer they are together. This tendency leads to clusters of similar values. In this context, spatial dependence is observed in both highlighted areas in Bronx and Queens Counties, where tracts with high

concentrations of the other races group and low household income levels are clustered together and surrounded by tracts and neighborhoods with similar variable values that gradually transition to extreme opposites. These patterns might imply that these neighborhoods share environmental or socio-economic factors that reinforce the distribution of lower-income levels within communities of the other races group. Recognizing this spatial dependence aids in understanding how socioeconomic conditions and racial demographics interrelate across adjacent neighborhoods, offering insights into broader trends shaping these communities.

New York's spatial interaction can be described as a composite model, where elements of monocentric (focused on a primary center) and polycentric (multiple centers of activity) patterns coexist. For instance, while Manhattan serves as the economic and cultural core (monocentric), other neighborhoods like Jackson Heights in Queens and Fordham in the Bronx act as self-contained centers of community life, employment, and cultural activities (polycentric elements). This interaction builds the flow of resources and opportunities that shape neighborhood dynamics and economic conditions, impacting where communities with higher other races population may experience varying access to income-generating opportunities.

The layout of New York City aligns with the multiple nuclei theory model, where various centers of economic and social activities emerge independently within the different counties, rather than being concentrated in a single core. This pattern is visible in the Bronx and in Queens, where clusters of high other races populations with low median household income are not only shaped by proximity to resources and services but might be influenced by cultural hubs, and local economies.

This analysis of New York City’s "Other races" population in relation to median household income uncovers significant clusters of socioeconomic disparity. The bivariate map reveals high concentrations of the target population with low income levels in neighborhoods like Morris Heights, Longwood, and University Heights in the Bronx, as well as smaller, more dispersed areas in South Corona and Jackson Heights in Queens. By using a corner model, the map distinctly highlights these extreme areas, though distinguishing mid-range values posed challenges due to subtle color differences in the intermediate range. These findings highlight spatial patterns of income disparity within racially diverse neighborhoods, inviting further analysis into the factors that drive these inequalities and how they intersect with demographic and geographic variables.